Kyle Seelman

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EDUCATION

University of Maryland, College Park, MD Aug. 2020 - Dec. 2025
Ph.D., Computer Science
Advisor: Jordan Boyd-Graber and Hal Daumé III
Thesis: *Improving Knowledge Discovery: Advanced Topic and Language Models*

Clemson University, Clemson, SC
B.S. Applied and Computational Mathematical Sciences Aug. 2016 - Dec. 2019
Magna Cum Laude
Undergraduate thesis: *Multilevel Support Vector Machines*

WORK EXPERIENCE

Roblox Sept. 2025-Present
Senior Machine Learning Researcher
- Built an extensible agentic platform combining multi-modal understanding (code, images, web) with tool-using agents for analysis, querying, and automation over full-world game content.
- Developed distilled vision-language models for 3D data safety classification, compressing large multi-modal architectures into efficient student models while preserving accuracy on complex 3D scenes and objects for safety detection.
- RL-trained interactive agents for Roblox gameplay, aligning behavior to world understanding and evaluation objectives via reward design and RLHF. Enabling large-scale gameplay agent testing and post-training iteration on tool use and environment interaction

Applied Research Laboratory for Intelligence and Security
Researcher Aug. 2023 - Dec. 2024
- Developed methods for predicting psychological dispositions from text using LLMs, combining prompt-based approaches with fine-tuning via LoRA and reinforcement learning from human feedback (RLHF).

SELECTED PROJECTS

Autonomous Biomedical Research Stack
https://github.com/kyleseelman

• Built an integrated pipeline for computational biology discovery: daily literature scanner (PubMed/bioRxiv, LLM relevance scoring) → persistent biomedical knowledge engine (SQLite graph + FAISS hybrid retrieval, multi-source ingestion) → multi-cycle research agent (literature-grounded hypotheses, 39+ tools, provenance, strategy review).

• Knowledge engine: entity resolution across Entrez/UniProt/DOID; plugin ingestion (Hetionet, STRING, GEO, agent sessions); graph/vector/hybrid query API.

• Research agent: autonomous hypothesis–experiment–analysis loops with persistent memory, Bayesian belief updates, and code-as-action execution; feeds from literature scanner.

PUBLICATIONS

Caption Overlap on OpenShape: A Stratified Audit and Counterfactual Retraining on Objaverse-LVIS
Harish Wajjala, **Kyle Seelman**
Submitted BMVC 2026

Archivist: Incorporating the World Knowledge of Neural Language Models into Topic Models as a Bayesian Prior
Kyle Seelman, Jordan Boyd-Graber
Submitted EMNLP 2025

From Text to Traits: Zero-shot Personality Facet Prediction with Open-source Language Models
Kyle Seelman, Anton Rytting, Triet Lee, Jordan Boyd-Graber
Submitted CoNLL 2025

Labeled Interactive Neural Topic Models: No Longer Take It or Leave It
Kyle Seelman, Mozhi Zhang, Jordan Boyd-Graber
Submitted ACL 2025

Decoding Digital Discourse: An Observational Study using Multimodal Text and Image Machine Learning Models to Classify Sentiment, Hate, and Anti-Hate
Thu T. Nguyen, Xiaohe Yue, Heran Mane, **Kyle Seelman**, Penchala Sai Priya Mullaputi, Elizabeth Dennard, Amrutha Alibilli, Junaid S. Merchant, Shaniece Criss, Yulin Hswen, Quynh C. Nguyen
JIMR 2025

What's Different between Visual Question Answering for Machine "Understanding" Versus for Accessibility?
Yang Trista Cao*, **Kyle Seelman***, Kyungjun Lee*, Hal Daumé III
Best Theme Paper Award, AACL-IJCNLP 2022

Towards Automated Classification of Complex 4D-STEM Datasets.
B. Savitzky, S. Zeltmann, L. Hughes, **K. Seelman**, M. Janish, M. Schneider, C. Gopal, P. Herring, A. Minor, C. Ophus.
Microscopy and Microanalysis 2020 Proceedings

Second order time discretization for a coupled quasi-Newtonian fluid-poroelastic system
H. Lee, H. Kunwat, and **K. Seelman.**
International Journal for Numerical Methods in Fluids. 2020.

TEACHING EXPERIENCE

TA, CMSC828U: Algorithms in Machine Learning Guarantees and Analyses
TA, CMSC828W: Foundations of Deep Learning
TA, CMSSC723: Computational Linguistics

AWARDS AND HONORS

John Charles Harden Award- Top undergraduate junior in mathematical sciences
President's List- Five consecutive semesters of receiving a 4.0 GPA
Phi Beta Kappa- Top 5% of class

TECHNICAL SKILLS

Languages: Python, R, Java, C++, SQL, Ruby
Frameworks: PyTorch, HuggingFace, LangChain, Tensorflow
ML Skills LLM fine-tuning, Prompt engineering, Topic modeling, Machine Translation, Retrieval-Augmented Generation (RAG)
Web Tools: HTML, CSS, Flask

GRADUATE COURSES

• Ethical Machine Learning • Applied Mechanism Design for Social Good • Foundations of Deep Learning • Visual Learning and Recognition • Computational Geometry • Machine Learning • Scientific Computing • Advanced Numerical Optimization • Linear Models